

NHAM QUOC HUNG

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EDUCATION

University of Texas at Austin

Jan 24 to Dec 26

- Master's in Computer Science (Optimisation, Machine Learning, Deep Generative Models)

National University of Singapore

Aug 18 to May 22

- Bachelor of Computing in Computer Science with Honours (Distinction - Database Focus Area)
- **ASEAN Secondary Scholarship** (2013-2017) - **ASEAN Undergraduate Scholarship** (2018-2022)

SKILLS AND CERTIFICATIONS

- **Programming Languages:** Python | Java | Scala | Kotlin | SQL | Bash Scripting | C
- **Data & Analytics:** Spark | Flink | Kafka | BigQuery | PostgreSQL | MongoDB | Couchbase | Redis
- **DataOps:** GCP | AWS | dbt | Airflow | Kubernetes | Iceberg | Great Expectations | Terraform | ELK
- **Certificates:** AWS Certified Cloud Practitioner (2021) | First Place (Causation x Healthcare) in AlxImpact Student Case Competition by McKinsey (2022) | Couchbase Associate Developer (2023) | Certified Kubernetes Application Developer (2024)

WORK EXPERIENCES

Data Engineer, Torilab

Jul 25 to Present

- Designed and deployed 30+ BI data pipelines and analytical models on GCP Cloud Composer and BigQuery, serving 6 mobile applications featuring AI Avatar companions and delivering key financial metrics to company executives and business leaders on Looker dashboards
- Architected end-to-end data infrastructure on GCP, integrating internal backend systems and 8+ third-party platforms (Shopify, TikTok/Meta/Google Ads, Adjust, Amplitude, OneSignal, HubSpot, Dify) into a unified lakehouse using Airflow, Cloud Run and dbt. Ensured reliable cross-domain data delivery across product, development, finance, marketing, sales, CRM, and AI core teams
- Built data quality, metadata management and governance foundations with AWS AI-DLC workflows to enable an agentic BI system capable of supporting self-service analytics

Software Engineer, Rakuten Asia

Jul 23 to Jul 25

- Managed Ad Item Master under the Global Ad Tech Supervisory Department - a mission-critical service which stores, delivers, tracks and reports revenue-generating ads for hundreds of millions of users and merchants on Rakuten Ichiba, one of Japan's largest online retail marketplaces
- Administered highly available Couchbase Enterprise clusters and Hive data warehouses serving as the source of truth for 500M+ ad items, shops, and campaigns across multiple data centres, enabling real-time ad delivery and downstream analytics for all ad platform users and accurately track user clicks and impressions for merchant charging
- Developed and maintained highly scalable, low-latency, cached, containerised and load-balanced backend systems, consumer/producer applications and APIs on top of Couchbase and Redis clusters using Spring Framework, Kubernetes, Kafka and Hive to serve super targeted ads

Data Engineer, Sea Limited

Jul 22 to Jul 23

- Developed batch and real-time data ingestion pipelines under Data Platform team, supporting HQ Finance department in data modelling and financial reporting on the NYSE for Shopee and Garena
- Monitored and maintained hundreds of terabyte-scale batch and streaming ingestion jobs daily across major titles (Free Fire, Call of Duty, Undawn) using Bash and Apache Spark
- Built a self-service web portal enabling non-technical Finance users to ingest data from major sources including relational databases, Google Sheets, CSV, and Kafka into HDFS — eliminating manual ticket requests and shortening data ingestion time from days to hours

Data Analyst Intern, Sea Limited**Nov 21 to May 22**

- Assisted Shopee Finance team in building efficient terrabyte-scale financial data models to report on the NYSE by researching into Spark SQL internals to optimise heavy Spark applications
- Built an auto-tuning program which can be attached to each data model to evaluate runtime statistics, identify potential bottlenecks and suggest optimised model configurations for data users
- Led optimisation efforts to identify critical logic and data issues such as skewness or improper null handling, reduced runtime and resource usage of certain applications by 10 to 50 percent and translated to significant model run time and infrastructure cost savings

Data Analytics and Machine Learning Intern, PSA International**May 21 to Nov 21**

- Assisted PSA Analytics team in researching on Reinforcement Learning to develop automated models aimed at reducing reliance on human operators when planning for costly port operations
- Prototyped a container pre-shuffling model utilising Proximal Policy Optimization algorithm to automatically pre-shuffle an existing container yard into its optimal arrangement within minutes. Minimized planning time and cost of subsequent container loading onto incoming vessels

COURSEWORK AND PROJECTS

Leadership and Mentorship**Jun 25 to Present**

- Mentored high school students and working professionals in honing academic, technical and soft skills to gain admission and obtain scholarships from top global QS-ranked universities
- Helped students gain admission into the following courses: Computer Science, Engineering and Business Analytics (NUS), Robotics (NTU), Computer Science and Design (SUTD), Master's in Computer Science (University of Texas at Austin), Master of Science in Computer Science (Georgia Institute of Technology), Applied AI (Singapore Institute of Technology), Business Informatics (Julius Maximilian University of Würzburg)

Master's Coursework, UT Austin**Jan 24 to Present**

- Completed in-depth coursework in the following areas: Advanced Linear Algebra for Computing, Online Learning and Optimisation, Deep Learning, Natural Language Processing, Advances in Deep Learning, Case Studies in Machine Learning, Android Programming
- Demonstrated expertise through projects, examinations and academic papers on the following Theory, Application and Systems coursework: writing rigorous mathematical proofs and implementing matrix decomposition algorithms in MATLAB, solving convex optimisation and multi-armed bandit problems, developing vision systems using Transformers and Convolutional Networks for SuperTuxKart - a racing simulator, building scalable language models and assessing the impact of dataset artifacts, optimising memory for distributed training of LLMs and VLMs, applying Deep Learning to demand forecasting problems, building practical mobile applications
- Sample: [Coursework Certifications](#)

Educational Sharing**Aug 22 to Present**

- Shared technical content and data engineering projects on platforms including WordPress and YouTube. Gained 2000 subscribers by covering topics such as building agentic workflows with GCP's Agent Starter Pack, setting up Big Data infrastructure for Hadoop, Spark, Yarn, Zeppelin on local and AWS environments, building fullstack websites with secure domain and routing, setting up streaming applications with Kafka and building data lakehouses with Apache Iceberg
- Sample: [Youtube Channel](#) and [Personal Blog](#)

AlxIMPACT Student Case Competition**Dec 21 to Feb 22**

- Participated in the Student Case Competition organised by QuantumBlack AI, McKinsey. Proposed an Intervention Recommendation model utilising simulation and time series prediction for policy makers to enact timely policies aimed at containing current and future pandemics. Presented to a panel of industry experts and secured First Place under the Causation x Healthcare track
- Sample: [AlxImpact Certificate](#)